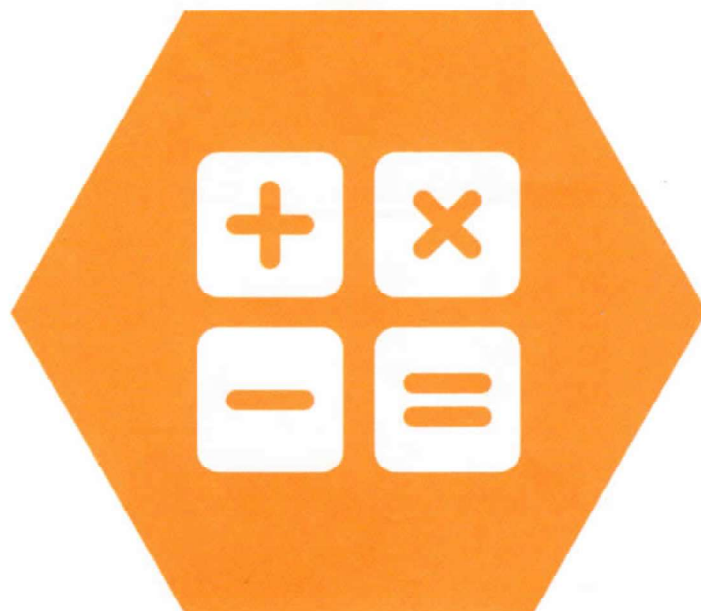




UNSW Global
THE UNIVERSITY OF NEW SOUTH WALES
SYDNEY • AUSTRALIA

PAPER A



2013 ICAS

International Competitions
and Assessments for Schools

MATHEMATICS

Educational Assessment Australia
eaa.unsw.edu.au

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

40 QUESTIONS

TIME ALLOWED: 45 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.
Use a 2B or B pencil.
Do **NOT** use a pen.
Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.
Your score will be the number of correct answers.
Marks are **NOT** deducted for incorrect answers.

There are **40 MULTIPLE-CHOICE QUESTIONS** (1–40).
Use the information provided to choose the **BEST** answer from the four possible options.
On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a ruler and spare paper.
You are **NOT** allowed to use a calculator.

1. How many candles are there on this cake?



- (A) 6
(B) 7
(C) 10
(D) 14

2. What is the mass of this clown?



- (A) three kilograms
(B) three and a half kilograms
(C) four kilograms
(D) four and a half kilograms

3. Raj counted backwards by twos to make this pattern:

38, 36, 34, 32, 30, ?

What is the next number in the pattern?

- (A) 32
(B) 31
(C) 29
(D) 28

4. This is a map of Edward's yard.



The  is north-west of the .

Which item is south of the .



(A)



(B)

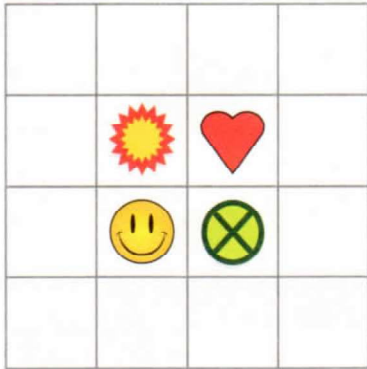


(C)



(D)

5. Mia has some pictures.



Which picture is in the third row from the top and second column from the left?



(A)



(B)

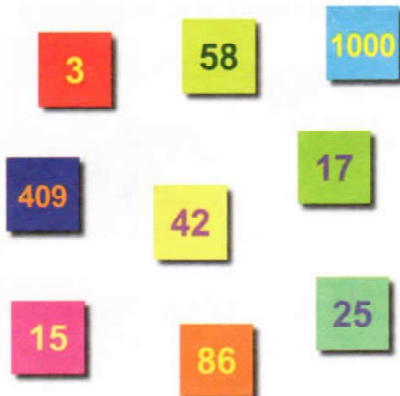


(C)



(D)

6. Raj has these number cards.



How many cards have an even number written on them?

- (A) 2
(B) 3
(C) 4
(D) 5

7. Anna created a number pattern. She started with the number 2 then added 3 each time to create the next number.

What are the first four numbers in Anna's pattern?

- (A) 2, 4, 6, 8
(B) 2, 5, 8, 11
(C) 2, 23, 233, 2333
(D) 2, 32, 332, 3332

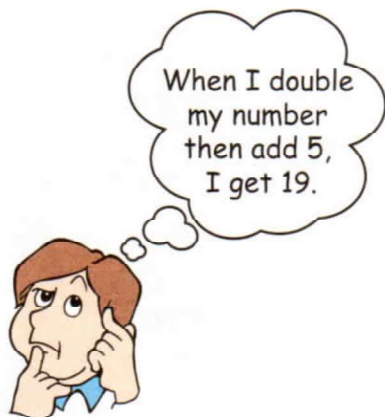
8. The spoon is about 10 centimetres long.



About how tall is the bear?

- (A) 10 centimetres
(B) 20 centimetres
(C) 50 centimetres
(D) 70 centimetres

9. Danny thought of a number.



What was Danny's number?

- (A) 7
- (B) 14
- (C) 43
- (D) 48

10. Which number is 4 less than 7091?

- (A) 7086
- (B) 7087
- (C) 7088
- (D) 7095

11. Sue drew a picture of a snail on her computer.



Next, Sue did something to the picture to make it look like this.



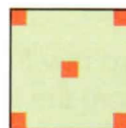
What did Sue do to the first picture to get the second picture?

- (A) She slid it.
- (B) She flipped it.
- (C) She turned it clockwise.
- (D) She turned it anticlockwise.

12. What is the number halfway between 20 and 46?

- (A) 26
- (B) 30
- (C) 33
- (D) 36

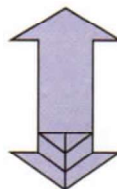
13. Which of these drawings has **exactly** two lines of symmetry?



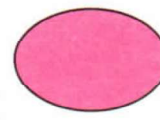
(A)



(B)



(C)



(D)

14. Male bees live for about 45 days. Female bees live for about two months.



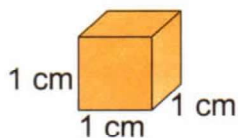
About how much longer than male bees do female bees live?

- (A) 15 days
- (B) 15 months
- (C) 43 days
- (D) 43 months

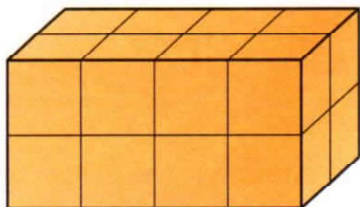
15. $23 + 1 = 13 + ?$

- (A) 1
- (B) 11
- (C) 24
- (D) 37

16. This is a cube.



How many of these cubes make up this solid?



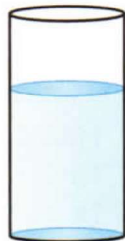
- (A) 12
- (B) 16
- (C) 20
- (D) 48

17. Sue poured one litre of water into each of the containers shown.

Which container has a capacity of less than one litre?



(A)



(B)



(C)



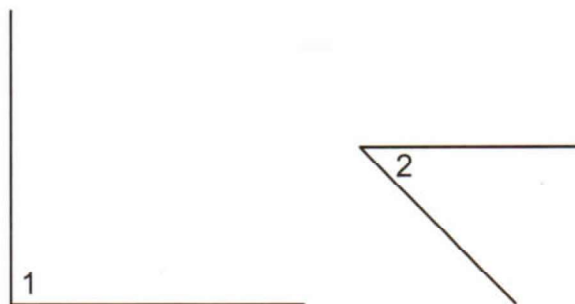
(D)

18. Half of Sam's pocket money is \$12.

How much is Sam's pocket money?

- (A) \$2
- (B) \$6
- (C) \$12
- (D) \$24

19. Which two angles are the same size?



- (A) 1 and 3
- (B) 1 and 4
- (C) 2 and 3
- (D) 2 and 4

20. Mia's mother made a pizza. Mia shared the pizza with her two sisters.



Mia's two sisters had equal amounts. Mia had as much as the two of them together. There was no pizza left.

How much of the pizza did Mia have?

- (A) half
- (B) one-third
- (C) one-quarter
- (D) three-quarters

21. There are 50 students in Year 3.

The table shows the number of Year 3 boys and girls in each sport group. Each student is in only one group.

Sport group	Girls	Boys
Red	3	9
Blue	8	7
Green	11	?

How many boys are in the Green group?

- (A) 38
- (B) 34
- (C) 16
- (D) 12

22. What is the value of **?** in the number sentence?

$$6 + \text{smiley face} = 15$$

$$6 \times \text{smiley face} = \text{?}$$

- (A) 48
- (B) 54
- (C) 60
- (D) 63

23. Sue has a number line drawn on the floor. Sue is making a pattern by jumping on the number line.

She starts at 3. She jumps three numbers forward to land on 6 then one number backward to land on 5.

She continues to jump three numbers forward followed by one number backward.

The numbers she landed on were:

3 6 5 8 7 10 9 12

On what number will Sue land next?

- (A) 11
- (B) 13
- (C) 14
- (D) 15

24. Edward has a bag with two yellow lollies and five red lollies.

Edward is going to take some lollies out of the bag without looking.

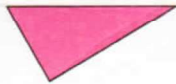
What is the smallest number of lollies that Edward must take out to be certain of getting a yellow one?

- (A) 2
- (B) 3
- (C) 6
- (D) 7

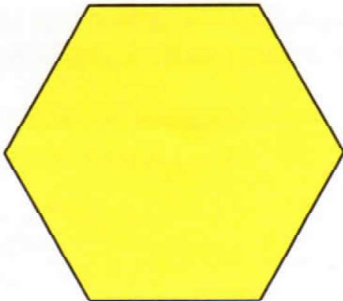
25. $603 - 75 =$?

- (A) 528
- (B) 538
- (C) 628
- (D) 672

26. Anna has tiles like this.



She wants to cover the hexagon with these tiles without gaps or overlap.



How many tiles does Anna need?

- (A) 14
- (B) 12
- (C) 10
- (D) 8

27. At an athletics carnival, every student entered at least one event.

The table shows the number of students in each event.

Event	Boys	Girls
Sprint	5	10
Long jump	10	5
High jump	2	3
Discus	3	2

What is the smallest number of students who were at the athletics carnival?

- (A) 5
- (B) 15
- (C) 20
- (D) 40

28. Tony and Anish tried out their toy cars to see which would go the longest distance.

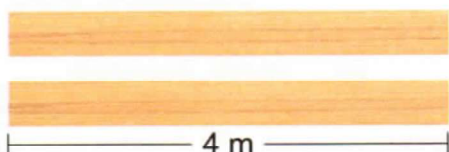


Tony's car travelled five metres. Anish's car travelled 189 centimetres.

How much further than Anish's car did Tony's car travel?

- (A) 489 cm
- (B) 411 cm
- (C) 311 cm
- (D) 184 cm

29. Jess wants to build a house for her pet. She has two pieces of wood. Each piece is four metres long.

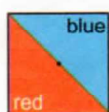


She needs to cut these pieces into shorter ones that are each 600 mm long.

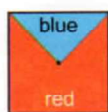
What is the largest number of 600 mm pieces Jess can cut?

- (A) 14
- (B) 12
- (C) 7
- (D) 6

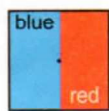
30. Four children each made a square out of cardboard. All the squares had the same area. Each child used two pieces of cardboard, one red and one blue.



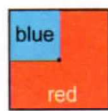
Anna



Mia



Raj

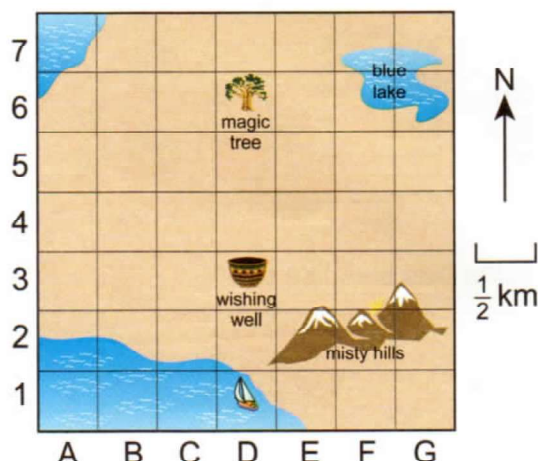


Danny

Which statement is true?

- (A) Raj's red piece has the same area as Danny's red piece.
- (B) Mia's blue piece has the same area as Anna's blue piece.
- (C) Anna's red piece has the same area as Danny's red piece.
- (D) Mia's blue piece has the same area as Danny's blue piece.

31. Raj has a treasure map.

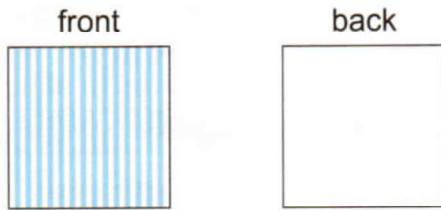


From the wishing well, he walks half a kilometre east, then one kilometre north. He then walks one and a half kilometres west to where the treasure is buried.

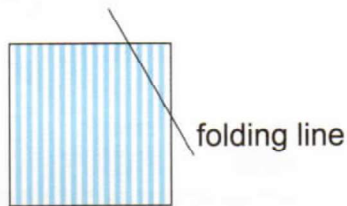
Where is the treasure buried?

- (A) B5
- (B) C4
- (C) E4
- (D) F5

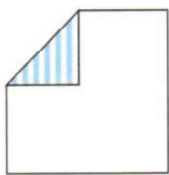
32. Sue has a piece of paper that has a pattern on the front, but is white on the back.



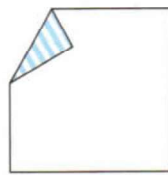
Sue folds a corner of the piece of paper towards the back.



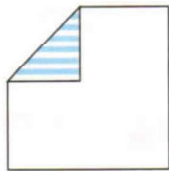
What would Sue's folded piece of paper look like from the back?



(A)



(B)



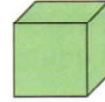
(C)



(D)

33. Danny has some cardboard squares that all have sides that are one metre long.

He made this closed box using six of his squares taped together. It is a cube that has a volume of one cubic metre.



Danny wants to make another closed box that has a volume of two cubic metres.

What is the smallest number of cardboard squares that he needs to make the other box?

- (A) 6
- (B) 8
- (C) 10
- (D) 12

34. Edward has four cats.



Cato



Fluffy



Ginger



Snowball

Cato is older than Snowball. Fluffy is younger than both Cato and Ginger, but older than Snowball. Cato is younger than Ginger.

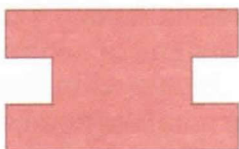
Which cat is the oldest?

- (A) Cato
- (B) Fluffy
- (C) Ginger
- (D) Snowball

35. Sue has a rectangle made of cardboard.



Sue cuts out a square from two sides as shown. Each square has a side one centimetre long.



Compared to the first shape, the perimeter of the new shape is

- (A) longer by four centimetres.
- (B) longer by six centimetres.
- (C) shorter by two centimetres.
- (D) the same length.

37. In Anna's school outstanding students win medals. For each medal they win, their class gets some points.

The table shows the number of points for each type of medal.

Medal	Points
 GOLD	3
 SILVER	2

Anna's class got 19 points.

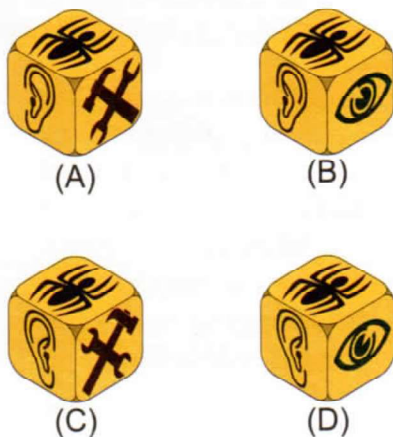
What is the maximum number of medals won by students in Anna's class?

- (A) 7
- (B) 8
- (C) 9
- (D) 19

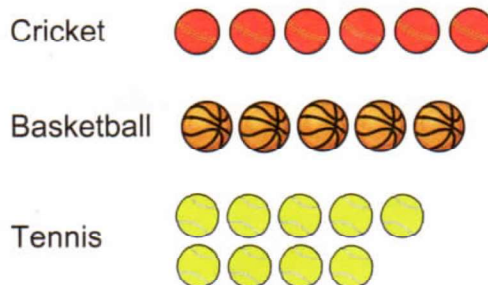
36. The picture shows views of a cube from three directions.



Which of these options shows the view of the cube from the west?



38. The graph shows the number of people in a group playing different sports. Each ball in the graph represents one person that plays the sport.



There are 14 people in this group. Each person plays at least one sport. Only one person plays all three sports.

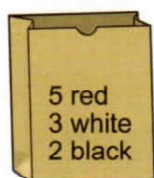
How many people play exactly two sports?

- (A) 4
- (B) 5
- (C) 6
- (D) 7

39. Danny has four bags of marbles.

Danny wants to pick a white marble without looking.

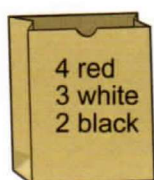
From which bag must Danny take a marble to have the greatest chance of picking a white one?



(A)



(B)



(C)

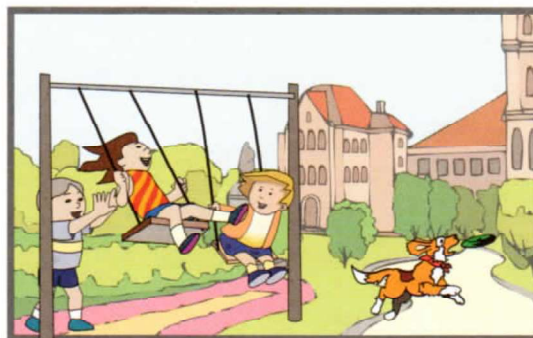


(D)

40. Anna, Raj and Edward agreed to meet at the park in the afternoon.

Edward left home at a quarter past one and arrived at a quarter to two.

Anna took 15 minutes more than Raj to get to the park, but 10 minutes less than Edward.



How long did it take Raj to get to the park?

- (A) 5 minutes
- (B) 20 minutes
- (C) 30 minutes
- (D) 35 minutes

HOW TO FILL OUT THIS SHEET:



- Rub out all mistakes completely.
- Print your details clearly in the boxes provided.
- Make sure you fill in only one oval in each column.

EXAMPLE 1: Debbie Bach

FIRST NAME

D	E	B	B	I	E
A	A	A	A	A	A
B	B	B	B	B	B
C	C	C	C	C	C
D	D	D	D	D	D
E	E	E	E	E	E

LAST NAME

B	A	C	H
A	A	A	A
B	B	B	B
C	C	C	C
D	D	D	D
E	E	E	E

EXAMPLE 2: Chan Ai Beng

FIRST NAME **LAST NAME**

C	H	A	N
A	A	A	A
B	B	B	B
C	C	C	C
D	D	D	D
E	E	E	E

A	I	B	E	N	G
A	A	A	A	A	A
B	B	B	B	B	B
C	C	C	C	C	C
D	D	D	D	D	D
E	E	E	E	E	E

EXAMPLE 3: Jamal bin Abas

FIRST NAME

J	A	M	A	L		B	I	N
()	()	()	()	()	()	()	()	()
(A)	(B)	(C)	(D)	(E)	(F)	(G)	(H)	(I)
(J)	(K)	(L)	(M)	(N)	(O)	(P)	(Q)	(R)
(S)	(T)	(U)	(V)	(W)	(X)	(Y)	(Z)	()
()	()	()	()	()	()	()	()	()

LAST NAME

A	B	A	S
()	()	()	()
(A)	(B)	(C)	(D)
(E)	(F)	(G)	(H)
(I)	(J)	(K)	(L)
(M)	(N)	(O)	(P)
(Q)	(R)	(S)	(T)
(U)	(V)	(W)	(X)
(Y)	(Z)	()	()
()	()	()	()

FIRST NAME to appear on certificate

[illegible]

LAST NAME to appear on certificate

[illegible]

Are you male or female?

☐ Male ☐ Female



Does anyone in your home usually speak a language other than English?

☐ Yes ☐ No

School name: _____

Town / suburb: _____

Today's date: / / **Postcode:**

DATE OF BIRTH

Day		Month		Year	
0	0	0	0	0	0
1	1	1	1	1	1
2	2		2	2	2
3	3		3	3	3
4			4	4	4
5			5	5	5
6			6	6	6
7			7	7	7
8			8	8	8
9			9	9	9

CLASS
(optional)

	(A)	(K)
	(B)	(L)
	(C)	(M)
	(D)	(N)
	(E)	(O)
	(F)	(P)
	(G)	(Q)
	(H)	(R)
	(I)	(S)
	(J)	(T)

TO ANSWER THE QUESTIONS

MULTIPLE CHOICE

Questions 1 to 40

Example: $4 + 6 =$

- (A) 2
- (B) 9
- (C) 10
- (D) 24

The answer is 10, so fill in the oval **C**, as shown.

A **B** **C** **D**



START

1 **A** **B** **C** **D**

2 **A** **B** **C** **D**

3 **A** **B** **C** **D**

4 **A** **B** **C** **D**

5 **A** **B** **C** **D**

6 **A** **B** **C** **D**

7 **A** **B** **C** **D**

8 **A** **B** **C** **D**

9 **A** **B** **C** **D**

10 **A** **B** **C** **D**

11 **A** **B** **C** **D**

12 **A** **B** **C** **D**

13 **A** **B** **C** **D**

14 **A** **B** **C** **D**

15 **A** **B** **C** **D**

16 **A** **B** **C** **D**

17 **A** **B** **C** **D**

18 **A** **B** **C** **D**

19 **A** **B** **C** **D**

20 **A** **B** **C** **D**

21 **A** **B** **C** **D**

22 **A** **B** **C** **D**

23 **A** **B** **C** **D**

24 **A** **B** **C** **D**

25 **A** **B** **C** **D**

26 **A** **B** **C** **D**

27 **A** **B** **C** **D**

28 **A** **B** **C** **D**

29 **A** **B** **C** **D**

30 **A** **B** **C** **D**

31 **A** **B** **C** **D**

32 **A** **B** **C** **D**

33 **A** **B** **C** **D**

34 **A** **B** **C** **D**

35 **A** **B** **C** **D**

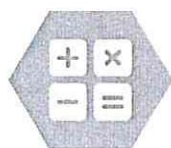
36 **A** **B** **C** **D**

37 **A** **B** **C** **D**

38 **A** **B** **C** **D**

39 **A** **B** **C** **D**

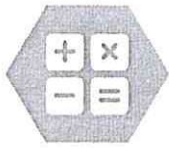
40 **A** **B** **C** **D**



M | 2013 Mathematics Answer Keys

Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Papers G & H	Papers I & J
1	B	B	A	C	D	B	A	C
2	B	B	B	A	B	D	C	C
3	D	C	D	A	B	C	C	B
4	D	D	A	C	D	B	B	C
5	C	C	D	C	C	A	D	D
6	C	C	B	D	A	A	B	A
7	B	A	A	D	D	D	A	D
8	C	A	A	A	D	B	A	A
9	A	C	B	C	C	C	B	A
10	B	A	D	B	B	D	D	B
11	B	B	C	D	D	A	A	C
12	C	D	C	A	A	C	D	D
13	D	D	B	D	D	A	C	A
14	A	A	D	A	A	A	A	C
15	B	C	B	C	A	C	D	B
16	B	B	B	A	A	D	B	C
17	D	D	A	A	B	A	D	D
18	D	C	C	B	A	D	D	B
19	B	C	C	A	A	D	C	D
20	A	B	B	A	C	B	B	A
21	D	C	D	A	A	C	B	B
22	B	D	C	B	C	D	C	D
23	A	D	A	D	D	B	B	C
24	C	A	C	B	D	C	C	D
25	A	D	B	A	A	B	C	B
26	B	C	B	B	D	B	C	A
27	C	B	D	C	A	A	D	A
28	C	C	D	D	C	C	D	B
29	B	C	A	B	B	A	B	C
30	D	A	B	A	B	B	C	C

(Please turn over)



M | 2013 Mathematics Answer Keys

Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Papers G & H	Papers I & J
31	A	B	B	D	A	C	C	A
32	D	B	A	B	D	C	B	B
33	C	C	C	C	C	B	B	D
34	C	A	D	D	C	B	D	A
35	A	B	C	C	A	C	C	C
36	A	C	D	12 012	950	6 006 06	6 006 06	10 010
37	C	D	C	128	301	98 098	4 004 04	4 004 04
38	A	A	B	90 090	135	10 010	216	351
39	D	A	B	126	12 012	90 090	127	104
40	A	B	C	350	600	16 016	18 018	65 065

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